Interpretable Card-Play Strategies from Evolved Logic Networks

A Neurosymbolic Agent for Sueca

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Most strong card-playing AIs are black boxes: can competitive play be human-auditable?

The status quo

Deep PIMC and neural policy nets.

Millions of floating-point weights → **opaque**.

"Why did you play that card?"
". . . the matrix says so."
". . . I saw 14 million futures, and this is the only one where we win."

Our goal

Evolved logic gates.

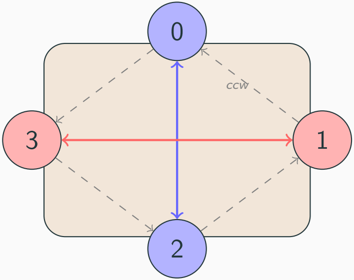
A handful of **human-readable IF/THEN rules**.

"I duck because the trick is cheap and I want to build a void."

Thesis: discrete logic gates + neuroevolution → competitive play *and* interpretability.

Sueca's distinctive rules create a rich strategic texture

Table layout *(4 players, 2 teams)*



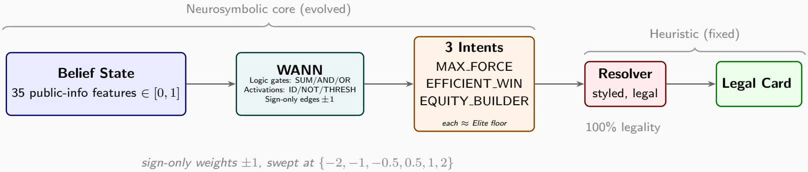
• Team 0&2 • Team 1&3

Key rules

- **40 cards** (no 8,9,10)
- Rank: **A** $>$ **7** $>$ **K** $>$ **J** $>$ **Q** $>$ **6–2**
- 7 = *manilha* (2nd-highest!)
- **120 points/deal**; game-point tiers
- Must **follow suit**; trump cuts
- **Public void tracking**
- Imperfect information: you see only your hand + history

These rules define the strategic vocabulary the WANN must learn.

The agent reads a public-information belief state through evolved logic gates and outputs a strategic intent



35 belief features
Hand, trick state, voids, boss detection, "can I win?"

3 intents
MAX_FORCE, EFFICIENT_WIN, EQUITY_BUILDER

Styled resolver
Every intent $\approx$ Elite; 100% legal outputs

Weight-agnostic networks replace learned weights with evolved logic-gate topology

Node types

Aggregation	Activation
SUM Σx_i	IDENTITY x
MIN = AND	NOT $1-x$
MAX = OR	THRESHOLD $x>0.5$

- No MEAN, no SIGMOID (interpretability)
- All outputs clamped to $[0, 1]$

Why it compiles to rules

- Connections: **sign only** (± 1)
- Shared weight sweep: $W \in \{-2, -1, -0.5, 0.5, 1, 2\}$
- Topology must work at *all* scales
- Every gate is a logic primitive $\rightarrow$ **Boolean-ish rules**
- Tie-breaking: **random** among max intents

*The architecture constraint **is** the interpretability guarantee.*

Three play intents: styled deviations around a strong Elite policy

MAX_FORCE

Elite + **control**.

When trump-long: lead a **low trump** to draw opponents' high trumps.

Otherwise: play **aggressively**.

"I have trump length — let me thin their trumps."

EFFICIENT_WIN

Exactly Elite.

Play the **cheapest** card that wins the trick.

If you can't win: cheapest legal sacrifice.

"Win cheap, lose cheap."

EQUITY_BUILDER

Elite + **tempo**.

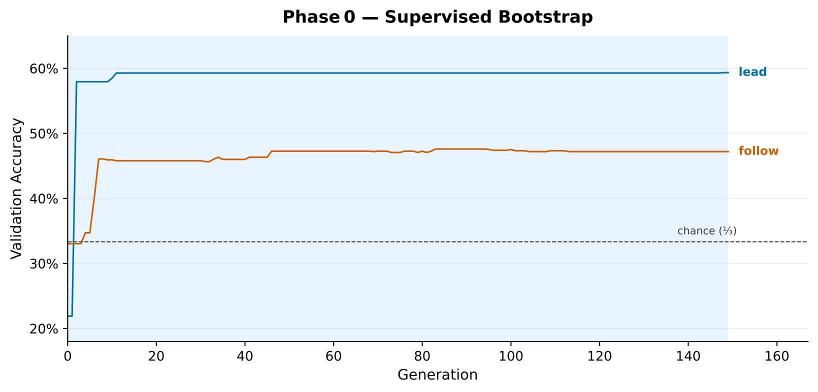
Lead **shortest** side-suit (build a void for later cuts).

Duck cheap tricks ($\leq$2 pts) when not last to play.

"I'll save my points and build void threats."

Each intent shares Elite's core; each benchmarks $\approx$Elite individually.
The WANN's job is to choose **when** to deviate, not **how** to play.

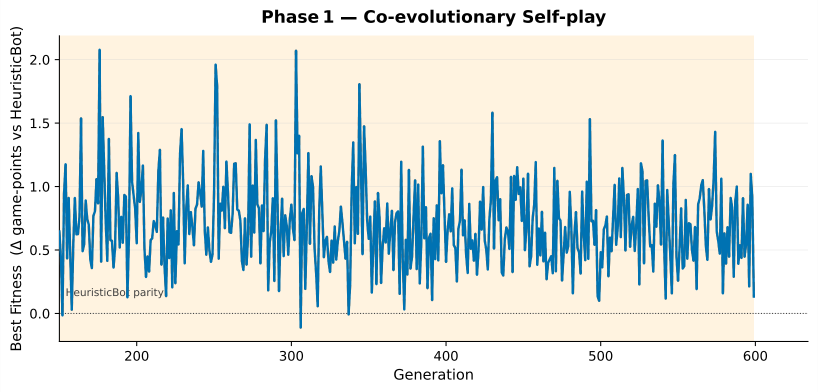
Phase 0: supervised bootstrap on PIMC expert data, starting from zero connections



PFS-NEAT from **zero** connections. Dataset split by `Am_I_Leading` into lead/follow.

Adaptive 2-stage mutation filter.

Phase 1: co-evolutionary self-play with dynamic per-decision brain routing



Lead and follow brains co-evolve. Dynamic routing via `Am_I_Leading` flag.

Delta-fitness with CRN. HOF + MAP-Elites opponent pools.

A cheap supra-Elite teacher made the champion 4.5× simpler at equal strength

Problem diagnosed

- Deep PIMC **only tied** Elite
- Root cause: **myopic leaf eval** (raw score, no positional estimate)
- Depth doesn't fix a blind leaf

Solution

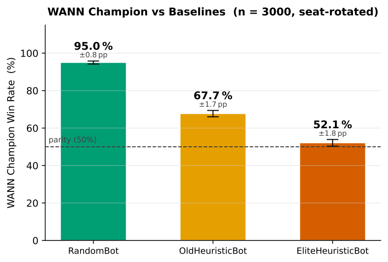
- Flat Monte-Carlo PIMC + **Elite playouts**
- Rollout policy improvement: 1-ply + Elite $\geq$ Elite
- **62% vs. Elite**, $\sim$1000× faster

What the better teacher bought

	Earlier	Final
vs. Elite	52.7%	**52.1%**
Hidden gates	132	**29**
Connections	188	**49**
Teacher	α-β	**rollout**

Statistically identical strength → 4.5× simpler.
The ceiling is the resolver, not the teacher.

The champion beats Elite 52.1% $\pm$ 1.8% ($n = 3000$)



Tournament matrix (win%, $n = 3000$)

	Rnd	Old	Elite	**WANN**
Random	50.0	10.8	4.7	5.1
Old	89.2	50.0	32.5	32.3
Elite	95.3	67.5	50.0	47.9
WANN	**95.0**	**67.7**	**52.1**	50.0

- 95% CI $[50.3\%, 53.9\%]$ **excludes 50%**
- Card points: 60.2 vs. 59.8
- WANN vs. Old: 67.7% (Elite: 67.5%)
- Prior best **30.2%**; first time the project beat Elite

Three root-cause fixes turned a collapsed network into an Elite-beating champion

Problem

Network collapsed to
“always EFFICIENT_WIN”
Intents unequally weak
20% / 37% / 25%

Fix #1

Styled resolver (Fix 1)
Every intent ≈ Elite
48% / 47% / 46%
Collapse → merely ties

Result: 52.7% vs. Elite
(132 gates, 188 conns)

Fix #2 & #3

Rollout teacher (Fix 2+3)
62% vs. Elite (itself)
→ 4.5× simpler champion

Result: 52.1% vs. Elite
(29 gates, 49 conns)

The follow brain compiles to 5 logic gates at depth 5 (verified behavior-preserving)

Compiled FOLLOW-brain rules (after constant-folding + alias inlining)

```
hidden_42 = NOT(Holds_Boss_Led)
hidden_48 = (1 + Any_Opp_Void_Led)
hidden_41 = NOT((1 + h_42 + 2*hidden_48))
hidden_40 = THRESHOLD((h_41 + Trump_Count) > 0.5)
hidden_55 = THRESHOLD(hidden_40 > 0.5)
MAX_FORCE      = 2*Game_Pts_Remaining
EFFICIENT_WIN  = 0
EQUITY_BUILDER = 1 + 2*hidden_41 + hidden_55
```

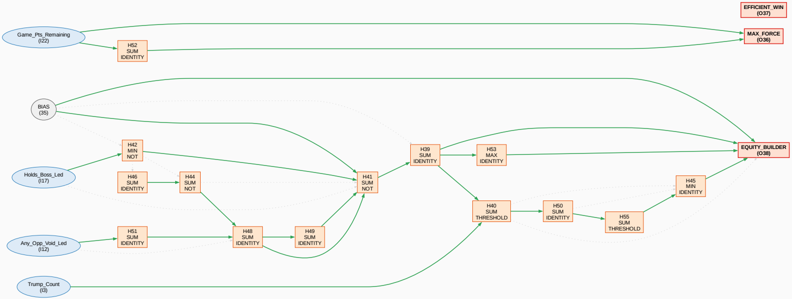
LEAD brain (10 gates, depth 6):

```
EQUITY = THRESHOLD(NOT(Trump_Count))
MAX_FORCE = (ahead & point-dense
             & partner-winning)
            OR (no short side suit)
```

EFFICIENT_WIN $= 0$ **in both brains:** the net never delegates to pure Elite; it always applies one of the two overlays.

Verified on 2000 random states: the removed symbols carry zero information.

Before folding: the raw evolved follow-brain topology



17 hidden gates, 44 connections. Constant-folding and alias inlining collapse the *active* logic to the **5 gates** on the previous slide, a 4.5× reduction versus the earlier champion.

Feature Space Analysis

	Pure intent	Champion	Oracle	RF predictor
vs. Elite	47–48%	**52.1%**	58%	63–71%

Pure intent: each of the 3 intents played alone (no WANN)
Champion: the evolved WANN choosing between them
Oracle: perfect intent selection at every decision
RF = Random Forest trained on belief features to predict the oracle's choice
$\rightarrow$ 63–71% shows the **features carry usable signal** the WANN has not yet learned.

What limits us, and what could raise the 58% ceiling

What limits us

- **3 intents $\approx$ Elite individually** creates a weak gradient the network converges to the simplest policy that ties Elite
- **Lead decision is harder**: Random Forest accuracy gap $+2.5$ pp (vs. $+7.9$ pp for follow)
- **Resolver-floor bottleneck**: realisable headroom above Elite is $\sim$8 pp

What could raise the cap

- **More-differentiated intents**: wider fitness gap $\rightarrow$ stronger learning signal
- **Opponent-card inference**: can a logic circuit learn to infer hidden cards from public information?
- **Knowledge distillation**: champion $\rightarrow$ smaller student, Pareto curve of strength vs. interpretability

Honest science: we measure and report our ceiling.

You can explore the champion's decision-making interactively in the browser

Sueca WANN: Network Inspector

Belief State (35 features)	Network Output
Has_Led_Suit = 1.0 Trump_Count = 0.4 Can_Beat_Winner = 0.0 Game_Pts_Remaining = 0.72 ...	MAX_FORCE = 0.12 **EQUITY_BUILDER = 0.87** EFFICIENT_WIN = 0.01 $\rightarrow$ Resolved: **Lead short side-suit**

React + WASM: Rust game engine compiled to WebAssembly
Play Sueca against the WANN, watch the logic flow in real time.
`sueca.mycsina.xyz`

Future Work & Conclusion

Future directions

1. **Richer intents**
   more-differentiated play styles
   $\rightarrow$ raise the oracle cap above 58%
2. **Opponent-card inference**
   can a logic circuit learn to infer hidden cards from presumed optimal play?
3. **Knowledge distillation**
   champion $\rightarrow$ even smaller student
   $\rightarrow$ Pareto curve of strength vs. size
4. **Collaboration**
   can a logic circuit display emergent collaboration capabilities in Sueca?

What we showed

- Evolved an **interpretable**, Elite-beating Sueca agent
- Verified pipeline:
  logic gates $\rightarrow$ **5 readable rules**
- Interactive website to explore the agent in a real game environment
- Fused PIMC with Elite rollout to estimate the true potential of any game position.
- All code, data, and rules are reproducible

Competitive, interpretable card-play AI is possible: the rules fit on one slide.

Thank you. Questions?

Key references

Gaier & Ha 2019	Weight-Agnostic Neural Networks
Stanley & Miikkulainen 2002	NEAT
Mouret & Clune 2015	MAP-Elites
Reisinger et al. 2004	Coevolving modular (L-NEAT)

Reproducibility

All commands in `README.md`

Canonical champion: `checkpoints/production/2026-06-14-2/`

`cargo build -p sueca_wann -release && cargo test -all`

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